

Network Forensics – Week 1 Report

Security Onion Hunt (PCAP Import) – MITRE ATT&CK; Mapping

Student: Jobayer Bhuiyan

Date: 2026-02-26

1. Objective

Import a provided PCAP into Security Onion and use the Hunt interface to identify Suricata alerts. From an alert event, extract MITRE ATT&CK; tactic and technique metadata and document the results with evidence screenshots.

2. Environment

- Platform: Security Onion web interface (Hunt).
- Data Source: Imported PCAP dataset (traffic capture).
- Primary Query: rule.severity = 2 (Suricata alert severity filter).

3. Method / Steps Performed

1. Imported the PCAP into Security Onion so events are indexed for search.
2. Adjusted the Hunt time range to match the PCAP time window (absolute range), then executed Hunt.
3. Searched Hunt using the filter rule.severity=2 to return relevant alert events.
4. Selected an alert from the Events table and expanded the event details.
5. Located MITRE ATT&CK; metadata fields in the alert: mitre_tactic_id and mitre_technique_id.
6. Documented the MITRE technique reference from the official ATT&CK; website.

4. Evidence (Screenshots)

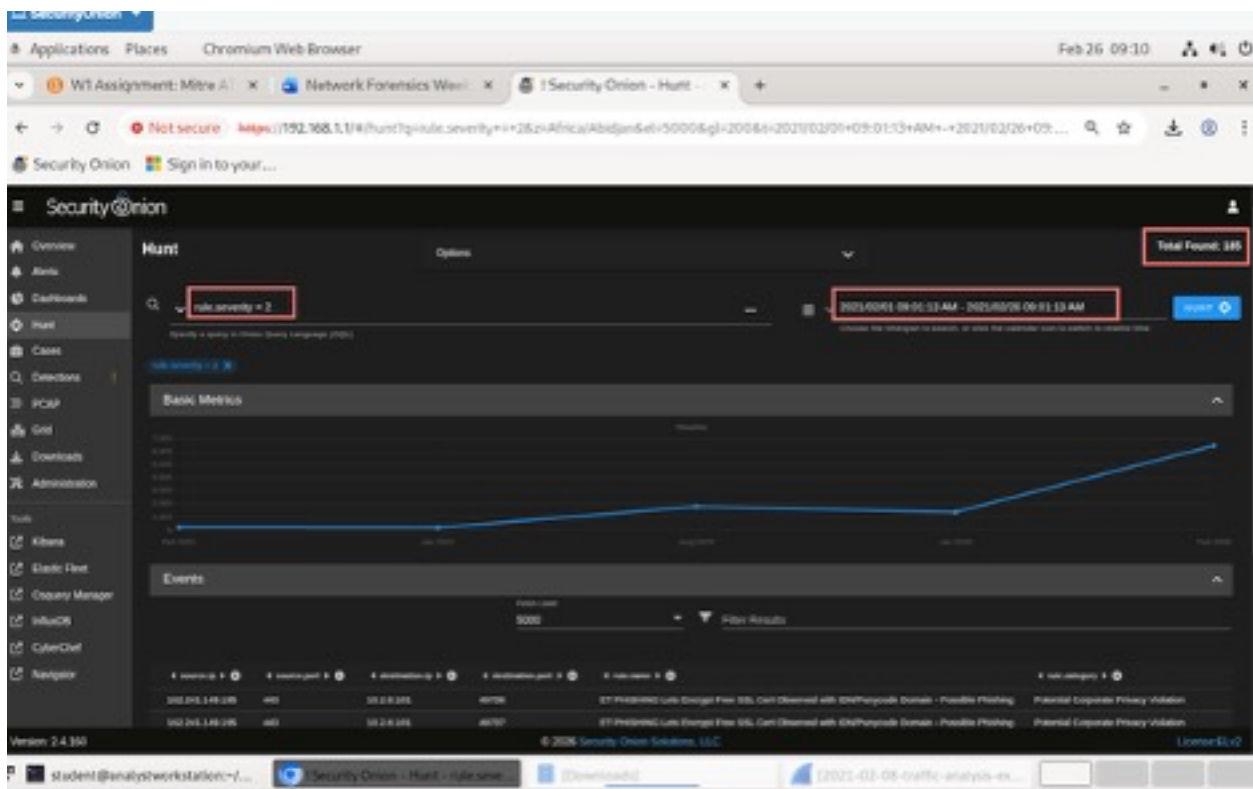


Figure 1. Hunt query using rule.severity=2 with an absolute time range; events returned.

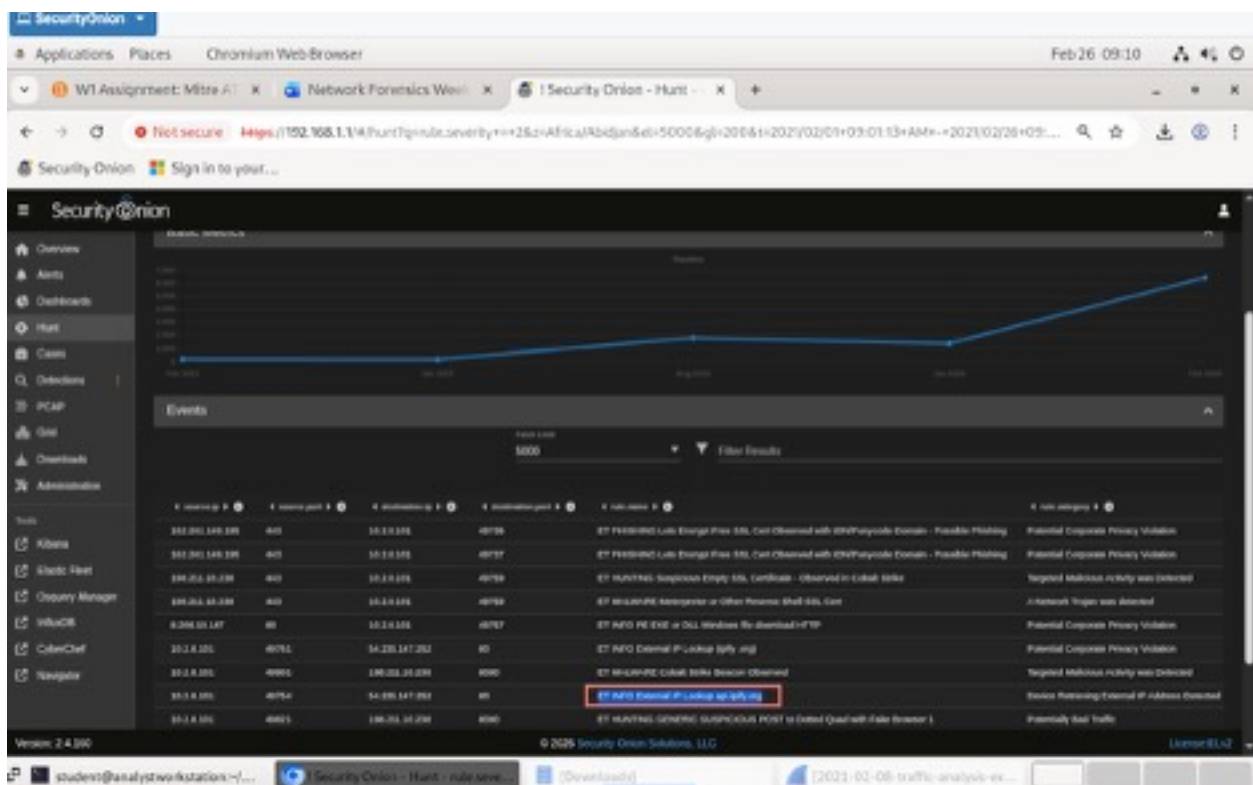


Figure 2. Events list showing Suricata alerts resulting from the severity filter.

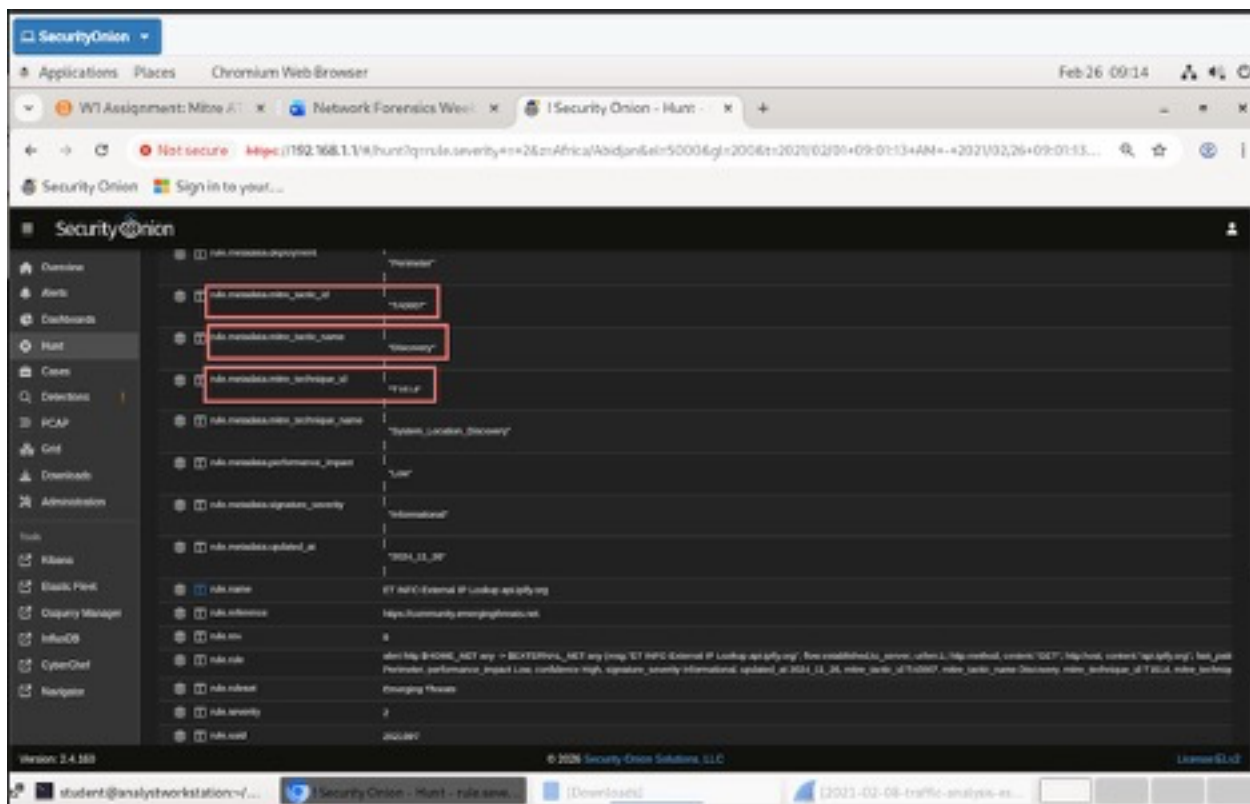


Figure 3. Expanded alert event details showing MITRE tactic and technique IDs.

5. Findings

Hunt Filter Used	rule.severity=2
MITRE Tactic ID	TA0007
MITRE Technique ID	T1614
Technique Name	System Location Discovery
MITRE Reference	https://attack.mitre.org/techniques/T1614/

6. Interpretation

The selected Suricata alert contains MITRE ATT&CK; metadata that maps the observed activity to Discovery (TA0007) and System Location Discovery (T1614). This technique is used when an actor attempts to determine location-related information about a host or environment (for example, by checking external IP/geolocation-related data).

7. Conclusion

The PCAP was successfully imported and searched in Security Onion. Using Hunt, severity-based filtering identified alert events, and an alert was mapped to MITRE ATT&CK; tactic TA0007 and technique T1614. Screenshots provide evidence of the search results and extracted metadata.

8. References

MITRE ATT&CK; – Technique T1614 (System Location Discovery): <https://attack.mitre.org/techniques/T1614/>